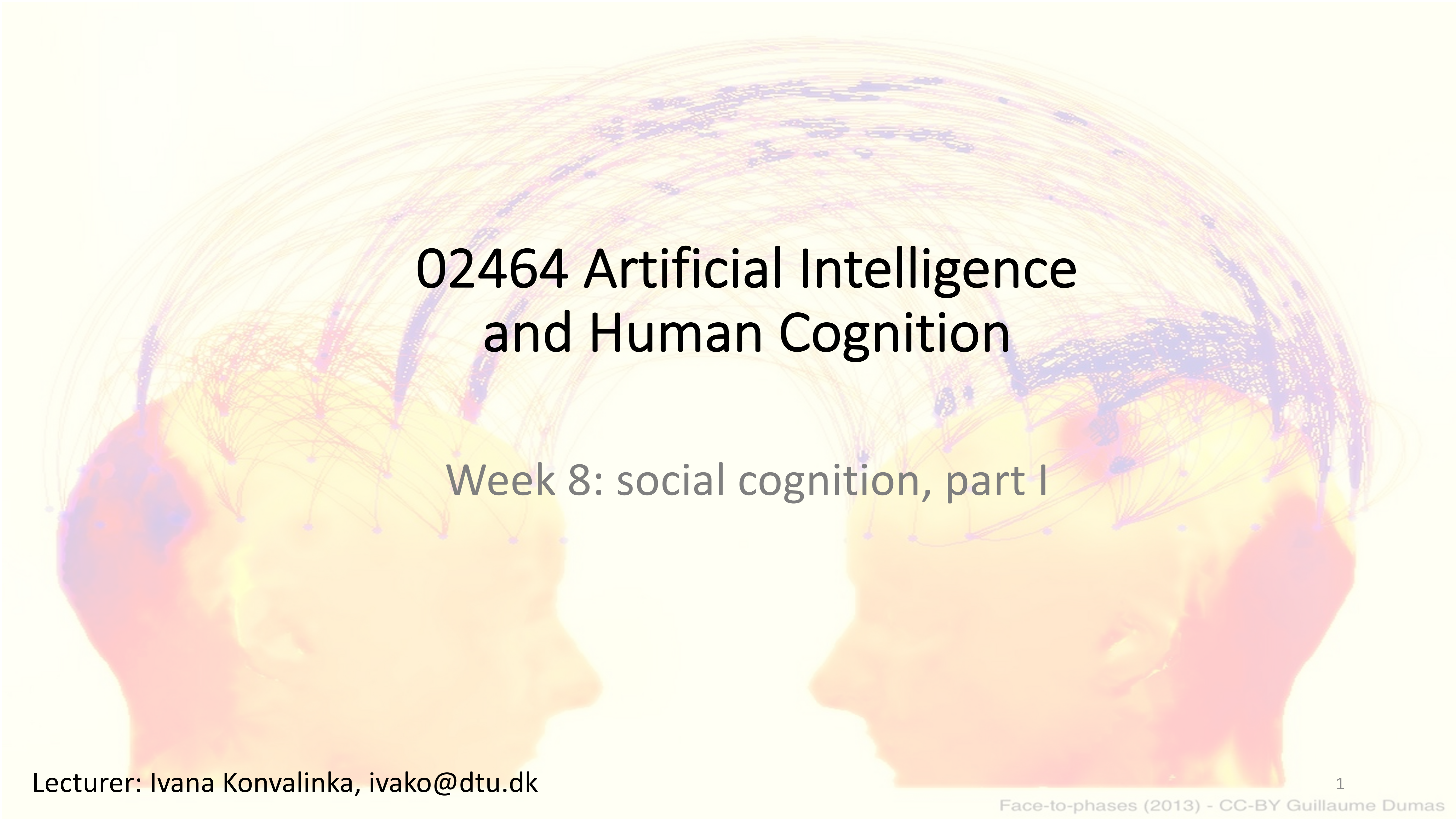
02464 Artificial Intelligence and Human Cognition

Week 8: social cognition, part I



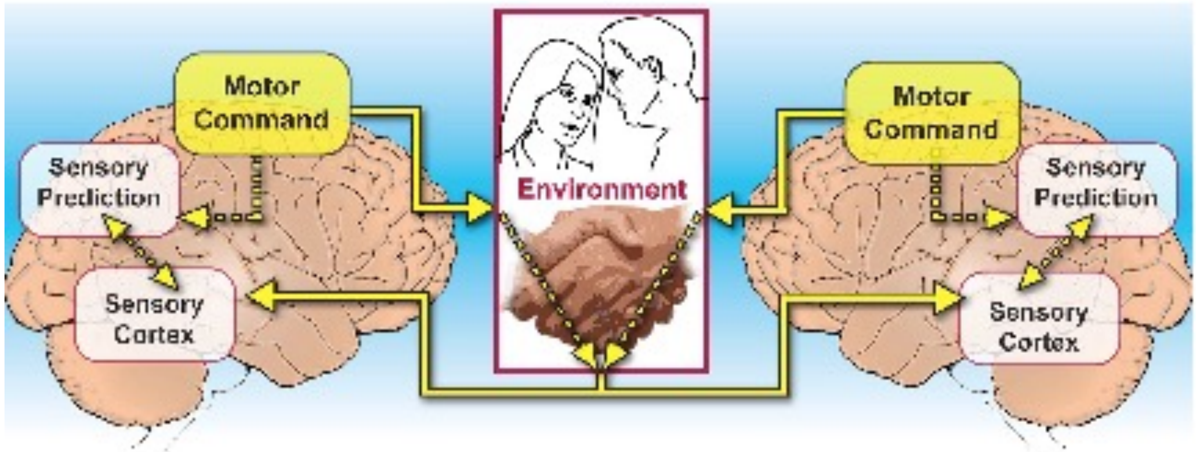
Lecturer: Ivana Konvalinka, ivako@dtu.dk

Social cognition (weeks 9 & 10) learning objectives

- You should be able to:
  - describe what social cognition is, what makes humans social, and why social cognition is important for AI
  - describe and differentiate between the two approaches to studying social cognition, including their advantages/disadvantages
  - describe mechanisms of social cognition, i.e., Theory of Mind, and how ToM is studied/measured in experiments
  - explain the concept and proposed function of the social brain, and be able to discuss this in the context of Bayesian inference
  - describe theories of how we understand others (Theory-Theory and Simulation Theory, mirror neurons)
  - describe mechanisms of joint action (e.g., spontaneous coordination, action prediction, adaptation, etc.)
  - describe when two heads are better than one (wisdom of the crowd, perceptual decision making experiment)
  - explain how we can use AI to measure neural mechanisms of social cognition (classification exercise)

Why social cognition is more complicated than motor control

+ infer high-level:
other people's
thoughts and goals



- we are faced with the same environmental uncertainty, sensory & motor noise, etc., but now we have also have to integrate the uncertainty of other people's actions, thoughts, and goals (and their uncertainty about us)

What makes us social?

- All animals are social. What makes humans more social?
- Humans have language
- Humans have created culture
  - We can learn even from people who are no longer alive
- Humans can consciously reflect on themselves and others
  - We have metacognition (awareness of one's own thought processes)
  - We have control (or delusion of control) over many of our social mechanisms

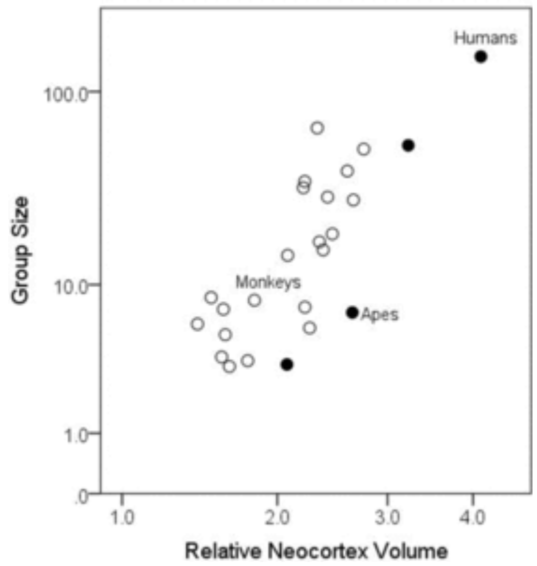
What "hardware" enables humans to be social?

- Human brain (neocortex) proportionally much larger in humans compared to other primates/mammals

- Neocortex
  - Outermost layer, largest part of cerebral cortex
  - comprises many brain areas involved in higher social cognition
    - language, empathy, emotion regulation, thinking about others

- Correlation between social group size among primates and relative neocortex volume: *social brain hypothesis*

- We are hardwired to interact with other people



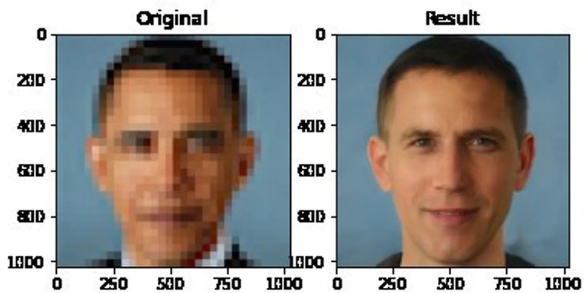
Social Cognition

- What is social cognition?
  - sum of all the processes that allow individuals to interact with one another
  - cognitive mechanisms underlying social behaviour and processing of social information

- Why is this important?
  - our brains are shaped through interactions with other people
    - how we think, feel, perceive, make decisions, remember
  - studies give insights into disorders of social cognition e.g., autism, schizophrenia

Why is understanding of social cognition mechanisms important for AI?

- Social robots and virtual assistants (to alleviate strain on healthcare systems)?
- Predicting human behaviour in real-time (i.e. autonomous vehicles)
- AI is developed by humans – that means it is embedded with people's biases and prejudices – understanding these mechanisms may help us overcome them or avoid transmitting them

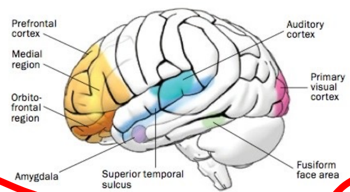


Social cognition mechanisms

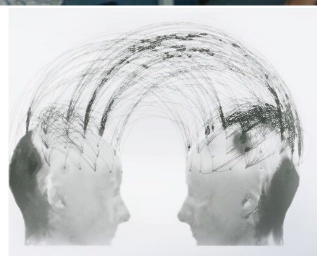
- **Prejudice**
  - widespread agreement among groups about what an untrustworthy face looks like, but no evidence that this is valid
  - derived from cultural norms and innate preferences for one's family and privileged groups rather than experience
  - children show preferences for lucky compared to unlucky groups
  - why do we have prejudice??
    - evolved for coping with situations where we have no prior knowledge

Approaches to studying social cognition

Individual in isolation
(individual approach)



Engaged in interaction
(joint action approach)



Automatic vs. Deliberate social mechanisms

- Most social processes are automatic, i.e. engaged without awareness
  - We make assumptions about people automatically based on their appearance (their emotions, trustworthiness, intention etc.)
  - We mirror each other's movements and mannerisms

- Higher-level social processes require consciousness
  - Increases possibilities for group actions
  - Uniquely human communication can emerge

Social cognition mechanisms

- Social learning: we learn by observing others
  - **gaze following** – tells us where another person is focusing their attention
  - **mirroring** – we covertly mirror other people's facial expressions: as a result, we experience the same emotions in ourselves
  - **prejudice vs. experience** – we learn to trust or distrust people through direct interaction
  - etc.

- Mentalizing/Theory of Mind

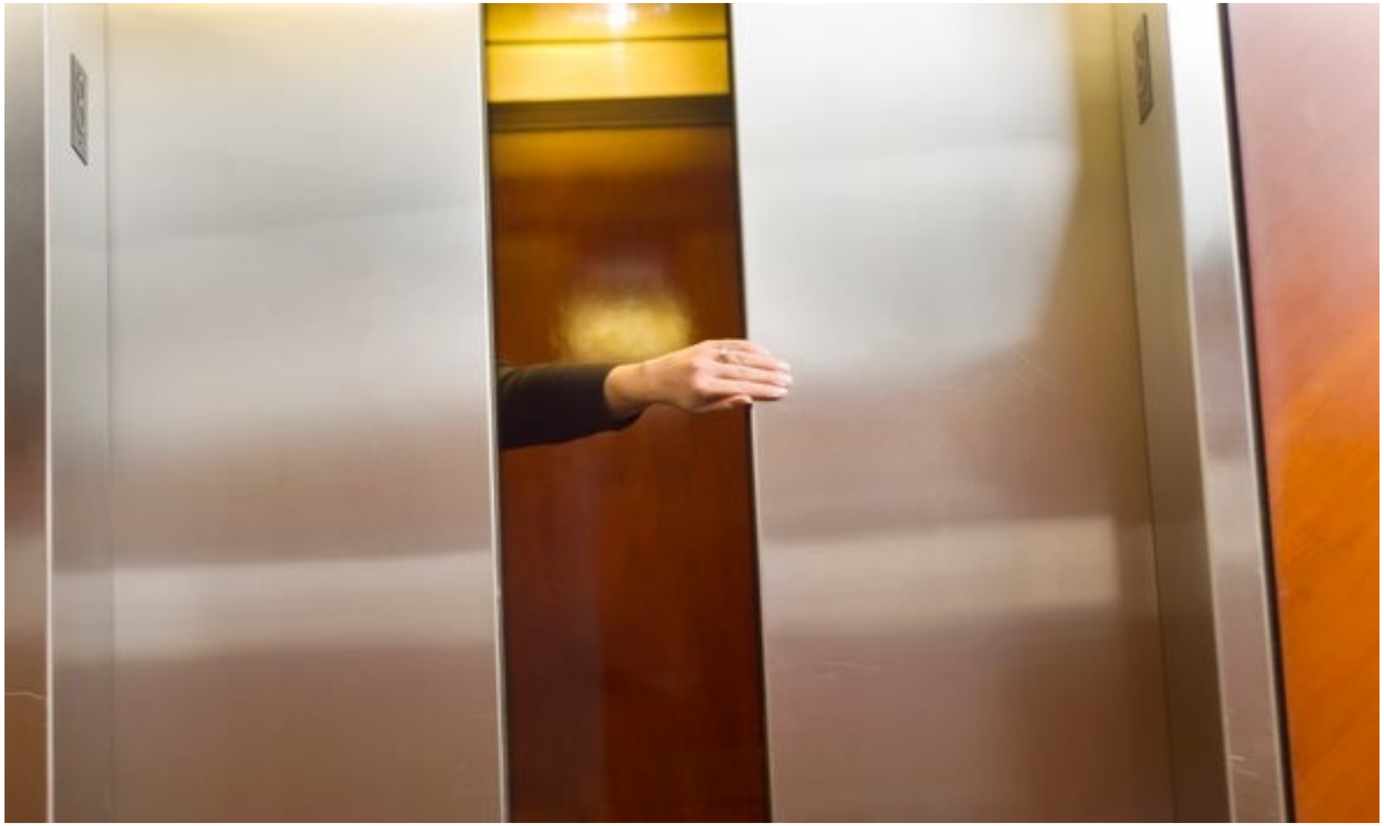
Mentalizing and Theory of Mind

- Mental states: desires, intentions
  - can be inferred from people’s movements and directions of their gaze

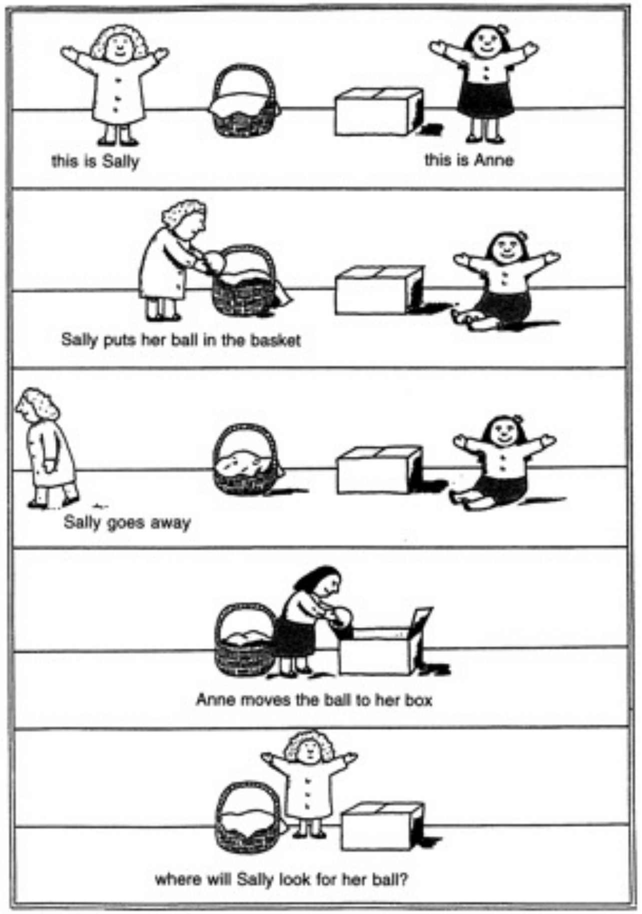
- **Mentalizing:** process of attributing and manipulating mental states
  - “reading others’ minds”
  - a prerequisite for being able to participate in a deliberately and consciously shared social world

- **Theory of Mind:** ability to attribute mental states, and predict and explain other people’s behaviour in terms of mental states (beliefs, desires, intentions, emotions)

Theory of Mind: anticipating needs and actions of others



Sally-Anne test: Explicit Theory of Mind



Measures a person's ability to attribute false beliefs to others

Typically developing kids pass this test at the age of 4-5

Children with autism typically do not, even up to their teens

Explicit vs. Implicit Theory of Mind

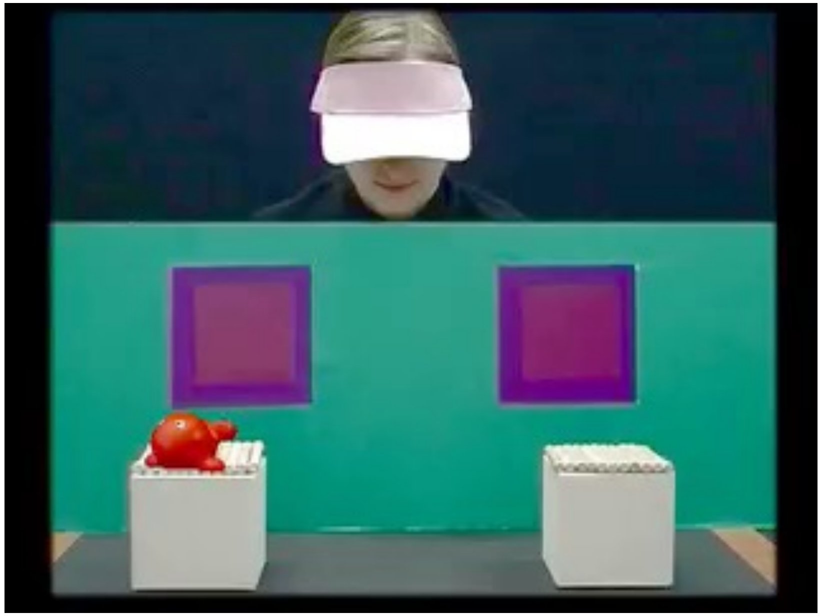
- Theory of Mind (ToM): ability to predict and explain other people's behaviour in terms of mental states (beliefs, desires, intentions)

- at the age of 4, children can pass the Sally-Anne task when **explicitly** asked to reflect about someone's mental states

- but children seem to develop **implicit** mentalizing earlier, by spontaneously attributing mental states to others

- implicit: without the need for language or executive processing

Measuring false-belief attribution in 2 year olds through action anticipation



17/20 babies gazed toward the correct window after illumination

Babies spent 2x as long focusing on the correct window as on the incorrect window overall

2 year olds correctly anticipate an actor’s actions by attributing a false belief to the actor

Eye-tracking used to measure looking behaviour

Southgate, Senju, Csibra (2007) *Psych Science*

Artificial Theory of Mind

- AI that learns from actions of others (through observation)

- ToMnet is a collective of 3 neural nets:
  - 1 learns the tendencies of other AIs based on their past actions
  - 2 forms a general concept of the current state of mind
  - 3 predicts the AI's actions based on the situation (with input from 1 & 2)

- struggles to predict in radically new environments (given its training context) or model actions of vastly different AI or human

Do LLMs have ToM?

Could you solve this problem for me? Ivana is playing with Annie. She takes Annie's ball and puts it into her bag. Annie gets upset and leaves for a bit. Ivana feels bad and moves Annie's ball back to Annie's bag. When Annie comes back, she wants her ball back. Where will she look for it?

This is a classic example of a false-belief test, which is used to assess theory of mind, or the ability to understand that others can hold beliefs different from one's own. In this scenario:

Annie leaves while the ball is in Ivana's bag, so her belief would be that the ball is still there because she didn't witness Ivana moving it back to her own bag. When Annie returns, she would likely look for the ball where she last saw it, in Ivana's bag.

This test shows that Annie would act based on her own perspective, unaware that the situation has changed in her absence.

Do LLMs have ToM?

- LLMs do in some cases exhibit humanlike responses on ToM tests, but does this mean that they have true understanding?
- These may be reliable proxies for assessing general abilities in humans, but not in AI systems
- Melanie Mitchell: we need new scientific methods to measure mechanisms of understanding

How do we study the brain basis of these social mechanisms?

- experimental tasks designed to engage social mechanisms (mirroring, mentalizing, etc.), as well as appropriate control conditions

- brain areas/mechanisms engaged in studied social mechanism:
  - brain areas “activated” during engagement in social task contrasted against brain areas activated during control condition

- brain mechanisms give more insight into social mechanisms

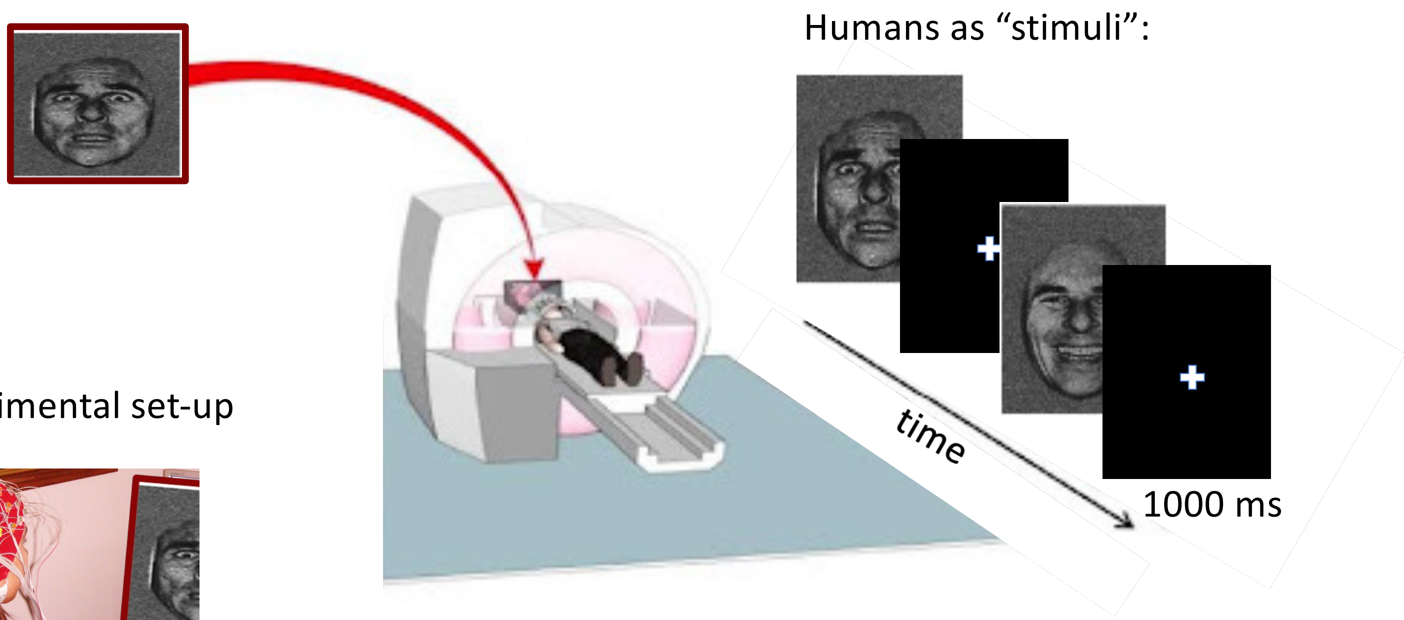
How does a brain give rise to social cognition?

- Brothers (1990) proposed that there is a restricted set of brain regions dedicated to social cognition

- Evidence comes from lesion studies in monkeys
  - Legions to the amygdala, orbital frontal cortex altered monkeys' social behaviour, made them socially isolated, etc.
  - This set of regions known as "the social brain"
  - Regions thought to be dedicated to social functions

Neuroimaging the social brain

fMRI experimental set-up



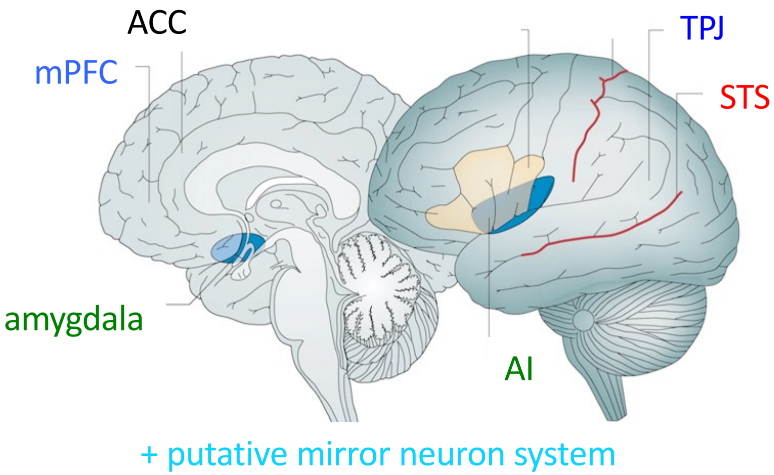
EEG experimental set-up

Tasks: 1st person perspective - observing pictures of faces expressing fear, disgust, rating trustworthy/untrustworthy faces, making decisions during economic games, observing actions, Sally-Anne type tasks, etc.

Adapted from Feldmanhall et al., 2012

The social brain

- Several brain regions attributed to socially relevant functions:
  - Observing faces and biological motion
  - Mentalizing
    - TPJ: immediate goals and desires
    - mPFC: more enduring traits of others
  - Emotional processes
  - Sharing experiences of others: many regions related to mirroring systems
  - Various functions (autonomic, emotion, decision making, empathy)



Adolphs 1999; Blakemore 2008; Frith 2007

Are “social brain” areas *only* social?

- amygdala is associated with processing of fearful social information, but also associated with positively/negatively valuable stimuli that are not social
- pSTS implicated in perception of people’s gaze direction, as well as processing of biological motion
  - function: to predict complex movement trajectories
- are the “social brain areas” only dedicated to social cognition?

The social brain

- Main function of the social brain (Frith, 2007)
  - **to make predictions during social interactions**
  - 2 problems the brain has to solve:
    - read mental state of the person interacting with based on their actions
    - make predictions about future behaviour on the basis of that state

- Can you formulate these problems as Bayesian inference? What is your prior, likelihood, and posterior?
  (Think back on motor control lecture and how we estimate whether a milk carton is empty or full. Replace context of the milk carton with the hidden intentions of another person.)

  Exercise (10 minutes)

Theories in understanding other minds

- How do we read other minds, despite their inaccessibility? i.e. how do we understand their actions, emotions, and intentions?

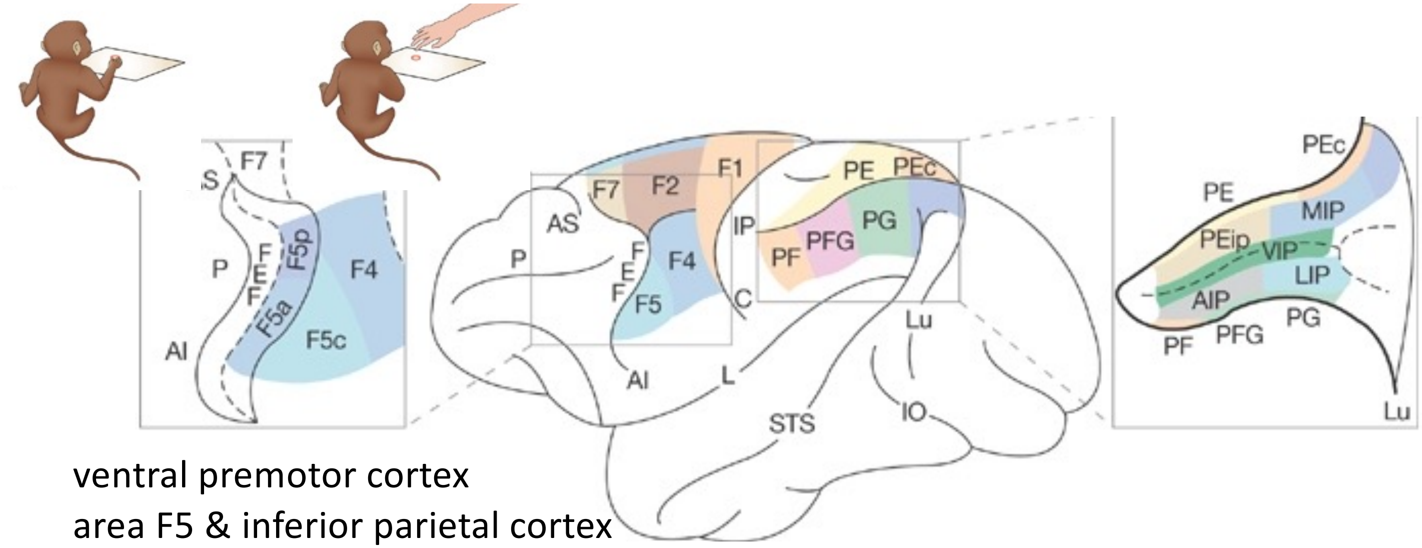
- **Theory-theory (TT)**
  - children innately develop theories about other people and their mental states, and try to verify them in reality (like scientists)
  - We hold a basic theory of psychology, which we use to infer mental states of others
  - We do not “experience” the mental state of others ourselves, but infer them based on our previous experiences
  - We understand and predict others’ mental states by inferentially applying a body of implicit knowledge/experience

Theories in understanding other minds

- **Simulation theory (ST)**
  - We simulate the actions or beliefs of others by activating the corresponding representation of those actions/beliefs within ourselves
  - We experience the mental states of others
  - We understand and predict others’ mental states by employing our own cognitive mechanisms to mentally model others’ cognitive processes (not by deploying a body of internally represented knoweldge structured like a theory)

Theories in understanding other minds

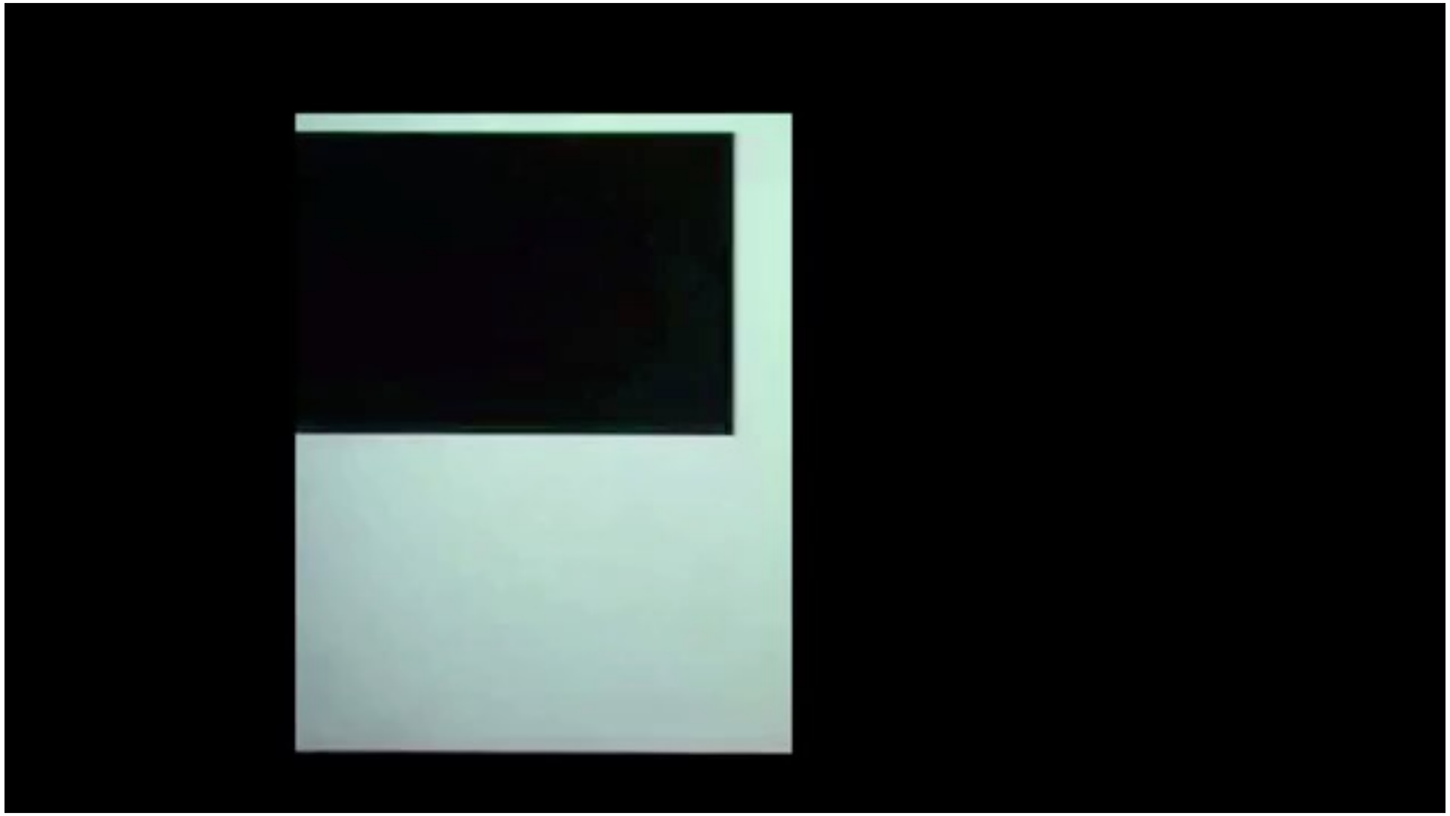
- Neuroscience provides support for Simulation Theory
  - Evidence comes from discovery of mirror neurons
    - neurons fire both when the monkey performs a motor act and observes another individual perform a similar act
    - supports the link between action observation and action execution
    - function? perception of motor actions and action understanding



ventral premotor cortex
area F5 & inferior parietal cortex

Rizzolatti et al., 2009

Mirror neurons



Mirror Neuron System (MNS)

- do mirror neurons exist in humans? still much debated

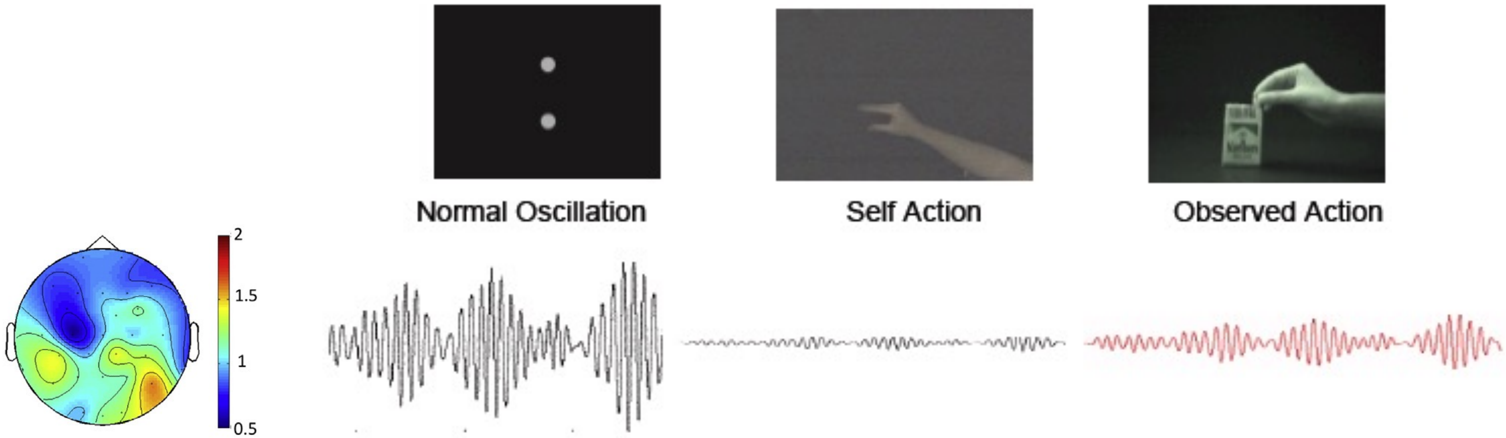
- individual neurons not found in humans, but engagement of corresponding brain areas – referred to as the mirror neuron system (using fMRI)
  - We do not know if the MNS brain areas contain mirror neurons

- activated by action execution, observation, imitation

- mirror neurons in monkeys fire only for goal-directed actions but in humans MNS has less strict constraints

EEG mu-rhythm: common coding of perception & action

- modulation of sensorimotor mu rhythm (10 Hz & 20 Hz) during self action and observed action
  - High amplitude during rest, suppresses (decrease in power) during active movements and observation of actions of another human



Human mirroring systems

- human mirror systems (NOT mirror neuron system!) are more extensive: activated by experience and observation of emotions, touch, auditory cues
  - “pain areas” activated when we see someone receiving a painful stimulus
  - somatosensory areas activated when we see someone else being touched

- location of activation depends on what is being observed

Human mirroring systems

- What is the function?
  - action understanding
  - intention understanding
  - automatic imitation
  - empathy
  - mentalizing (mirroring is the first step – e.g. another's emotion – providing the initial estimate of the mental state of the person we are interacting with?)

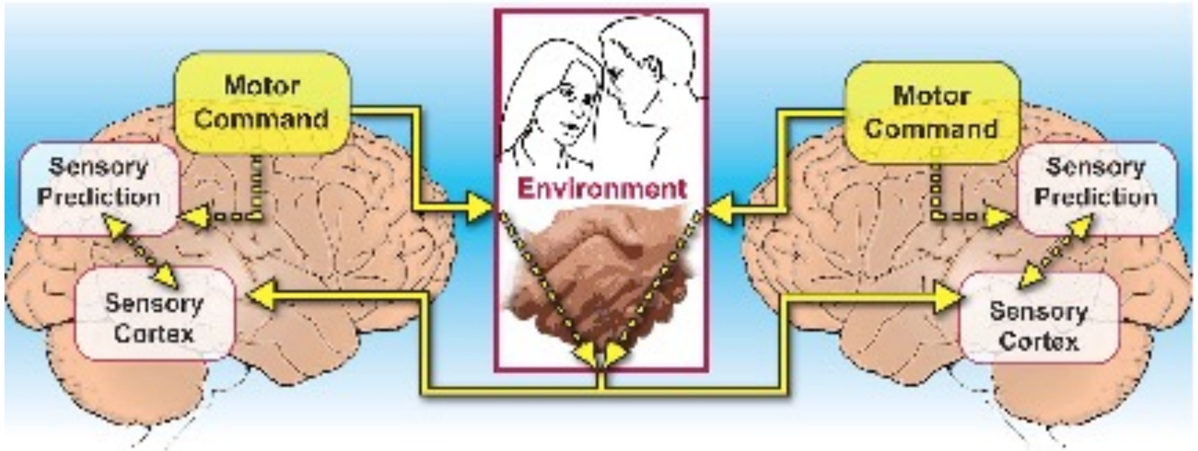
Problem with single-person neuroscience

- Many regions that are part of “the social brain” have numerous functions, which are not always related to the processing of social stimuli

- Social cognition is fundamentally different when people engage in reciprocal interaction with adaptive others

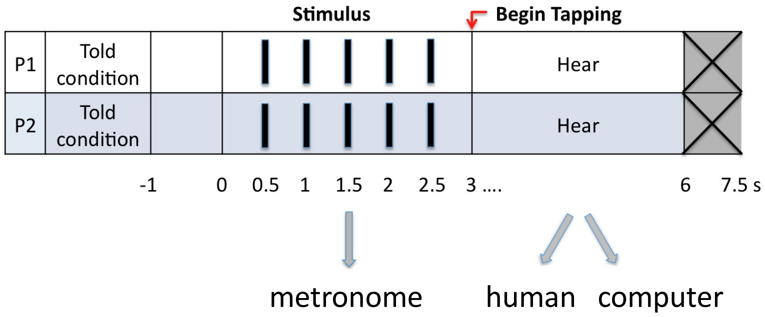
- Social interaction is a dynamic process, operating at various time-scales, which cannot be captured using static stimuli

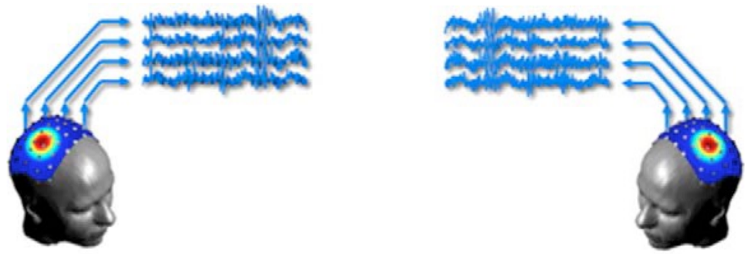
Joint action approach

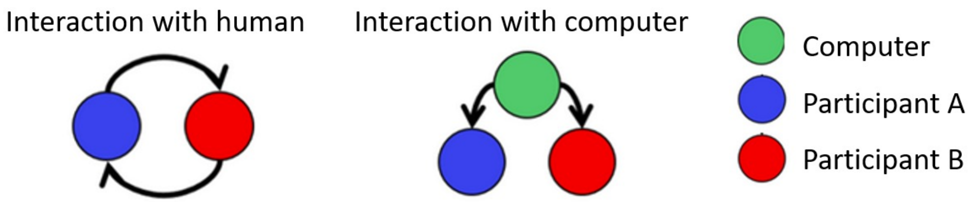


- perception and action in a social context
- one person's action output = another person's perceptual input (and vice-versa)
- joint action: any form of social interaction whereby two or more individuals coordinate their actions in space and time to bring about a change in the environment

Joint action study







- Task: tap to the given tempo, and synchronize with auditory feedback
- **EEG**: Simultaneous 32-channel EEG recorded from both members of a pair
- Goal: uncover brain mechanisms underlying social interaction (in contrast to non-social condition)?

AI & Human cognition

- How AI mimics human cognition
  - parallels between how AI performs vs. how the human brain/cognition performs
  - e.g., Tobias on intelligence, chess example: AI performs well at the high-level problem, but poorly at the low-level (moving like a human), etc.

- Applying AI to human data (e.g. from experiments) to better understand human cognition
  - e.g., applying machine learning models to make sense of the brain data

AI & Human cognition

- When (and why) is it useful to use a data-driven machine learning approach to analyze the e.g. brain data?
  - if we want to make predictions
  - if we don’t have a strong hypothesis (exploratory analysis)
  - to find patterns in the data, e.g., by data mining of large data sets
  - to better understand the brain, and mechanisms of human cognition (e.g., uncover general principles)

Exercise today & next week: classifying experimental conditions based on brain data

- Classification: logistic regression

- Classes: (1) interaction with human; and (0) interaction with computer condition

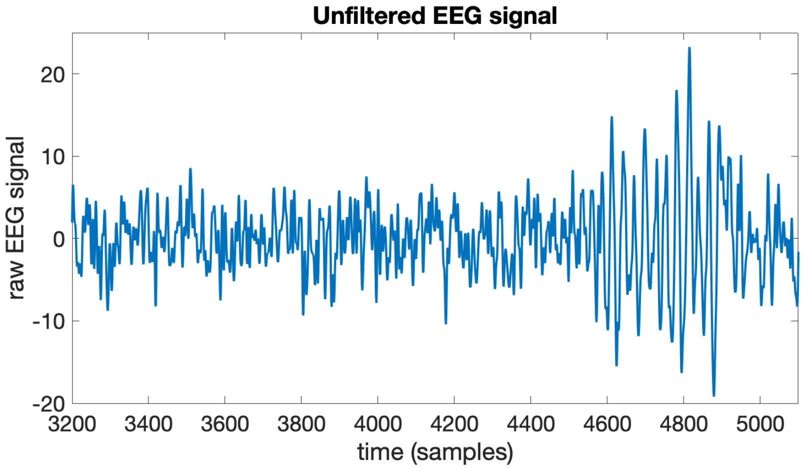
- Features: 10 Hz power during tapping instances/trials for all electrodes of member 1 and member 2 in a pair (30 electrodes x 2 members)
  - data from 4 pairs provided

- 10-fold cross-validation, Forward sequential feature selection
  - since shuffling is done differently for each run, run the SequentialFeatureSelector with random_state = 42 (that way, we can later check the results).

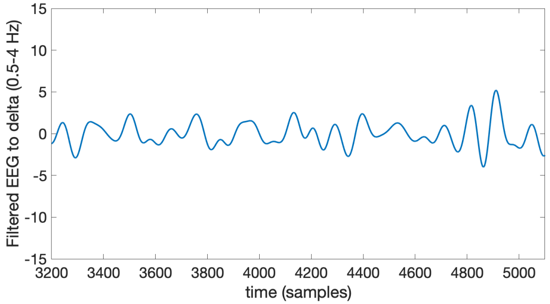
Exercise today & next week: classifying experimental conditions based on brain data

- Data: 4 pairs
  - Features: power of 10 Hz oscillations
    - 60 features (30 corresponding to person 1’s electrodes, 30 for person 2’s)
    - data structure: trials x features matrix
  - Labels, y:
    - human/computer labels: 1/0
    - data structure: trials x 1

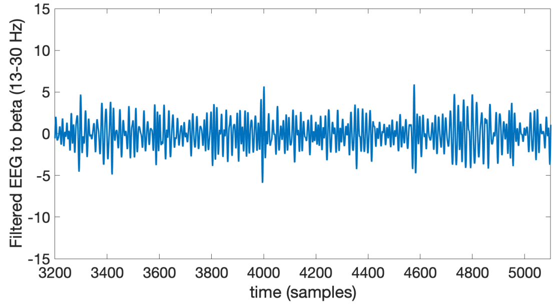
Features: 10 Hz power from filtered EEG data



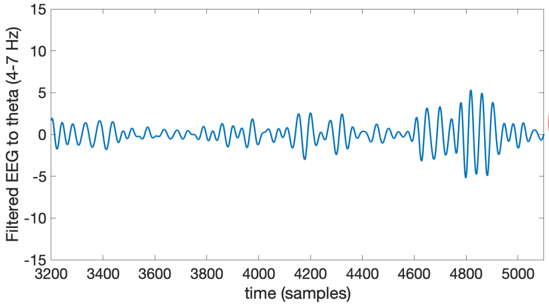
Delta (0.5-4 Hz)



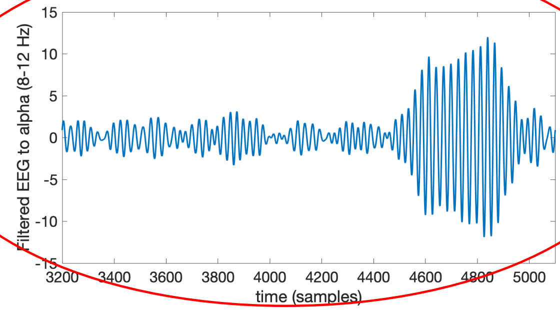
Beta (13-30 Hz)



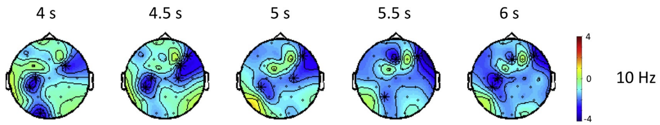
Theta (4-7 Hz)



Alpha (8-12 Hz)



Why choose power (amplitude squared) of 10 Hz oscillations as features?

Human vs. Computer interaction: 

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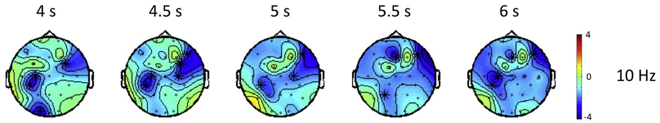

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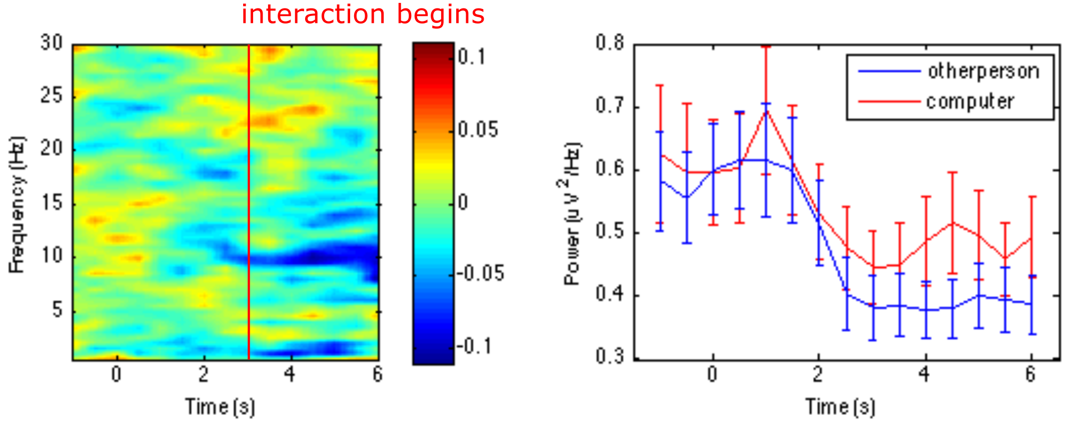
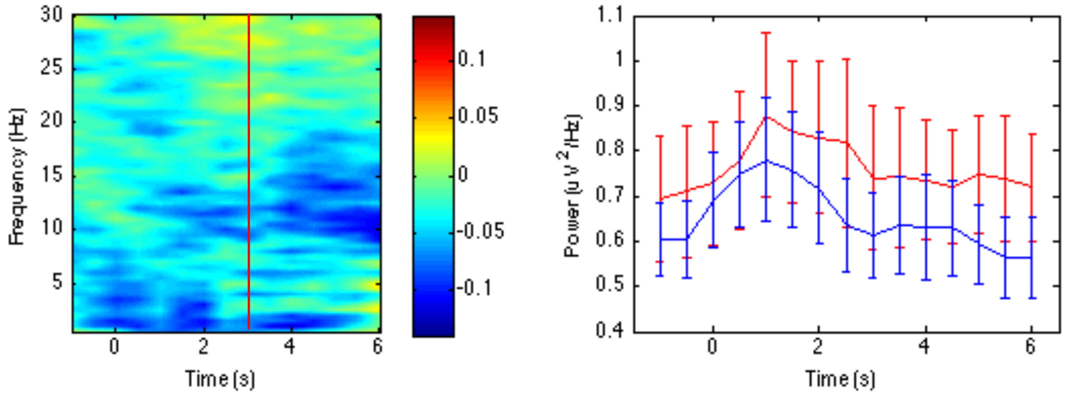


Mu-suppression
Activation (or release from inhibition) of **motor** cortex

Left-central/Sensorimotor Electrodes



Release from inhibition of **frontal** cortex

Frontal Electrodes

Emergence of leaders and followers

- A leader-follower relationship emerged across most pairs

- Leader: focuses on own movements more than other’s during human-human interaction (while still following computer in the non-human condition)

- Follower: focuses on other’s movements more than own (in both human-human and computer interaction)

- Here, we’re using both leader and follower data to classify human interaction condition and computer condition. **Which features do you expect to be the best predictors?**